

Out-of-Laboratory Longitudinal Gait Assessment of Participants Before and After Anterior Cruciate Ligament Reconstruction Surgery: An Observational Longitudinal Study

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Background: After anterior cruciate ligament reconstruction (ACLR), individuals often demonstrate altered knee kinematics during walking. Understanding these changes across recovery time points is important for informing rehabilitation and return-to-sport decisions.

Objectives: To evaluate the longitudinal changes in knee sagittal-plane kinematics before and after ACLR during varying walking speeds in nonlaboratory environments and to describe hip- and ankle-joint kinematics.

Design: Observational longitudinal study.

Setting: Hospital.

Patients or Other Participants: A total of 40 patients with ACLR (age = 23.9 ± 5.9 years, height = 1.76 ± 0.10 m, mass = 76.03 ± 13.4 kg) and 17 healthy matched controls (age = 28.4 ± 5.9 years, height = 1.70 ± 0.10 m, mass = 68.3 ± 15.4 kg) were recruited.

Main Outcome Measure(s): Knee-joint sagittal-plane kinematics were measured using 7 inertial measurement units preoperatively and 3 and 5 months postoperatively while participants walked at slow, normal, and fast speeds.

Results: Preoperatively, compared with the contralateral knee, the injured knee exhibited greater minimum flexion during normal and fast walking ($P \leq .008$) and less knee flexion at the first peak during fast walking ($P = .006$). Statistical parametric mapping indicated differences throughout the gait cycle

at all speeds ($P \leq .03$). Compared with the control knee, the injured knee had greater minimum flexion during normal and slow walking ($P \leq .03$). At 3 months postoperatively, compared with the contralateral knee, the injured knee exhibited greater minimum flexion across all speeds ($P \leq .005$) and less knee flexion at the first peak during fast walking ($P < .001$). Statistical parametric mapping indicated differences throughout the gait cycle at all speeds ($P \leq .03$). Compared with the control knee, the injured knee remained more flexed at the minimum angle across all speeds (all, $P < .001$) and exhibited less knee flexion at the first peak during slow walking ($P = .03$). At 5 months postoperatively, differences between limbs were reduced, with differences in minimum flexion at all speeds ($P \leq .03$). Statistical parametric mapping differences were limited to specific gait cycle portions during normal and fast walking ($P \leq .01$). Compared with the control knee, the injured knee remained more flexed at the minimum angle during slow and normal walking ($P \leq .005$). Lastly, hip adaptations resolved, and ankle asymmetries persisted during terminal stance.

Conclusion: Patients with ACLR demonstrated progressive improvements in knee sagittal-plane kinematics, indicating a recovery trend. However, the recovery was nonlinear across different walking speeds.

Key Words: walking, kinematic analysis, lower limb

Key Points

- Patients with anterior cruciate ligament reconstruction (ACLR) showed improved symmetry in knee-flexion angles at 3 and 5 months post-ACLR during various walking speeds.
- Asymmetries in knee, hip, and ankle kinematics were observed before ACLR, indicating altered movement patterns, and persisted post-ACLR.
- The nonlinear recovery process across different walking speeds highlights the need for assessing and incorporating varied walking speeds in rehabilitation protocols.

Anterior cruciate ligament (ACL) reconstruction (ACLR) often results in lasting kinematic asymmetries while walking, reduced quality of life, and increased risk of posttraumatic osteoarthritis (PTOA).^{1–3} Early gait assessment within the first 6 months post-ACLR is critical for guiding rehabilitation and reducing long-term joint degeneration risks, as these asymmetries can substantially affect knee function and potentially

accelerate PTOA development.^{2,4} Despite its clinical importance, comprehensive biomechanical assessment accounting for preoperative status, varying walking speeds, and hip and ankle compensatory adaptations remains underexplored. Moreover, traditional analyses focus primarily on discrete time points of the gait cycle, limiting our understanding of continuous movement patterns in nonlaboratory settings.

The role of preoperative biomechanical status is often underexplored, with few researchers providing preoperative and postoperative assessments.⁵ This gap is important, as preoperative biomechanics can influence postoperative recovery outcomes.^{6–8} Specifically, Büttner et al demonstrated that gait asymmetries are present preoperatively and persist through recovery time points at 2, 4, 6, and 12 months post-ACLR.⁹ Furthermore, Davis-Wilson et al observed bilateral gait abnormalities in patients at 6 and 12 months post-ACLR compared with uninjured controls, emphasizing the need to assess recovery across multiple stages and walking speeds.¹⁰

Walking speed influences lower extremity biomechanics post-ACLR, with faster speeds typically exacerbating biomechanical deficits.^{11–13} This is particularly important, as daily activities require walking at varying speeds—from slow ambulation during routine tasks to fast walking when commuting or exercising. By assessing these speed-dependent changes, clinicians can better understand how patients adapt their movement strategies in real-world contexts and can adjust rehabilitation interventions to improve performance across all activity levels. However, authors of only a handful of studies have assessed the influence of walking speed on gait kinematics post-ACLR.^{11–13} Understanding how the relationship between walking speed and biomechanics changes throughout rehabilitation might lead to better insight into recovery patterns and real-world mobility demands.

Although knee kinematics is a primary concern post-ACLR, assessing hip and ankle kinematics alongside it provides insight into compensatory strategies and overall lower limb adaptation. Researchers have shown that patients with ACLR display compensatory sagittal-plane adaptations, specifically reduced hip flexion and ankle dorsiflexion, contributing to asymmetrical gait patterns.^{14–16} These adaptations may contribute to altered joint-loading patterns throughout the kinetic chain.

Movement analysis is focused on specific gait events, such as minimum and maximum points, providing important snapshots of movement. However, continuous analysis methods such as statistical parametric mapping (SPM) offer a more comprehensive view of movement mechanics and can be used alongside traditional discrete point analysis.¹⁷ Specifically, SPM addresses these limitations of discrete analysis by examining continuous data, allowing the evaluation of how joint angles change over time and identifying differences or asymmetries that discrete analysis might miss. For example, SPM analysis of walking has revealed that knee sagittal-plane kinematic asymmetries post-ACLR occur during extended portions of the stance phase rather than just at discrete peak angles.^{11,18} Despite these advantages, SPM remains underused in gait analysis, particularly with inertial measurement unit (IMU) data in ACLR populations.¹⁹

Traditional optoelectrical camera systems, while accurate, are expensive and confined to laboratory environments, reducing ecological validity.^{20,21} In contrast, wearable technologies such as IMUs allow real-world kinematic data collection, although normative values are needed to interpret differences from optoelectrical data.²² Although no researchers have directly linked IMU-based gait analysis to ACLR and PTOA outcomes, IMU systems have shown promise in evaluating gait deviations associated with osteoarthritis (OA). When paired with SPM, wearable sensors have demonstrated the

ability to identify gait deviations associated with knee OA that are characterized by asymmetries in ankle, knee, and hip sagittal-plane kinematics during stance and swing phases.²³ Greater asymmetries were associated with lower quality-of-life scores measured using the Knee Injury and Osteoarthritis Outcome Score and Hip Disability and Osteoarthritis Outcome Score, suggesting that IMU-based gait analysis can provide meaningful insights into biomechanical asymmetries and related quality-of-life measures.

The primary purpose of our study was to evaluate knee sagittal-plane kinematic differences between injured and contralateral limbs and healthy controls pre-ACLR and at 3 and 5 months post-ACLR using an IMU system during slow, normal, and fast walking speeds. The secondary purpose was to describe potential compensatory adaptations of hip and ankle sagittal-plane kinematic differences at the same time points. We hypothesized that patients with ACLR would demonstrate differences in knee sagittal-plane kinematics between the injured and contralateral limbs and the injured and control limbs, with these differences decreasing but persisting by 5 months post-ACLR. We also hypothesized that faster walking speeds would magnify these between-limbs differences compared with normal or slow walking speeds.

METHODS

Design

Data for this study were obtained from a randomized clinical trial (ClinicalTrials.gov ID: NCT05001594), in which the effect of local muscle vibration on the lower limb kinematics of patients with ACLR was assessed. Following the framework of Lakens for sample-size justification in resource-constrained studies derived from previous trials, we performed a sensitivity power analysis for a sample of 40 participants.²⁴ This analysis revealed that, with an α level of .05, our study had 80% power to detect effect sizes (ESs) of ≥ 0.45 , 90% power for ESs of ≥ 0.53 , and 95% power for ESs of ≥ 0.58 in our between-limbs comparisons.

In accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines, we conducted a longitudinal observational study.²⁵ The study involved individuals scheduled for ACLR surgery, with assessments conducted preoperatively and at 3 and 5 months postoperatively. Gait kinematics at 3 different walking speeds were collected using an IMU motion-capture system (Xsens Technologies, BV).

Participants

Participants were recruited from the Rambam Health Care Campus Orthopaedic Department between 2020 and 2023 during their scheduled preoperative meetings. Gait analysis was performed at 3 time points: preoperatively and at 3- and 5-month postoperative follow-up visits.

We included male and female participants aged 18 to 40 years who were scheduled for ACLR using hamstrings, bone-patellar tendon-bone, or quadriceps graft. The surgeries took place between 2020 and 2023 and were conducted by a team of 5 experienced surgeons, each with 7 to 25 years of senior surgical experience.

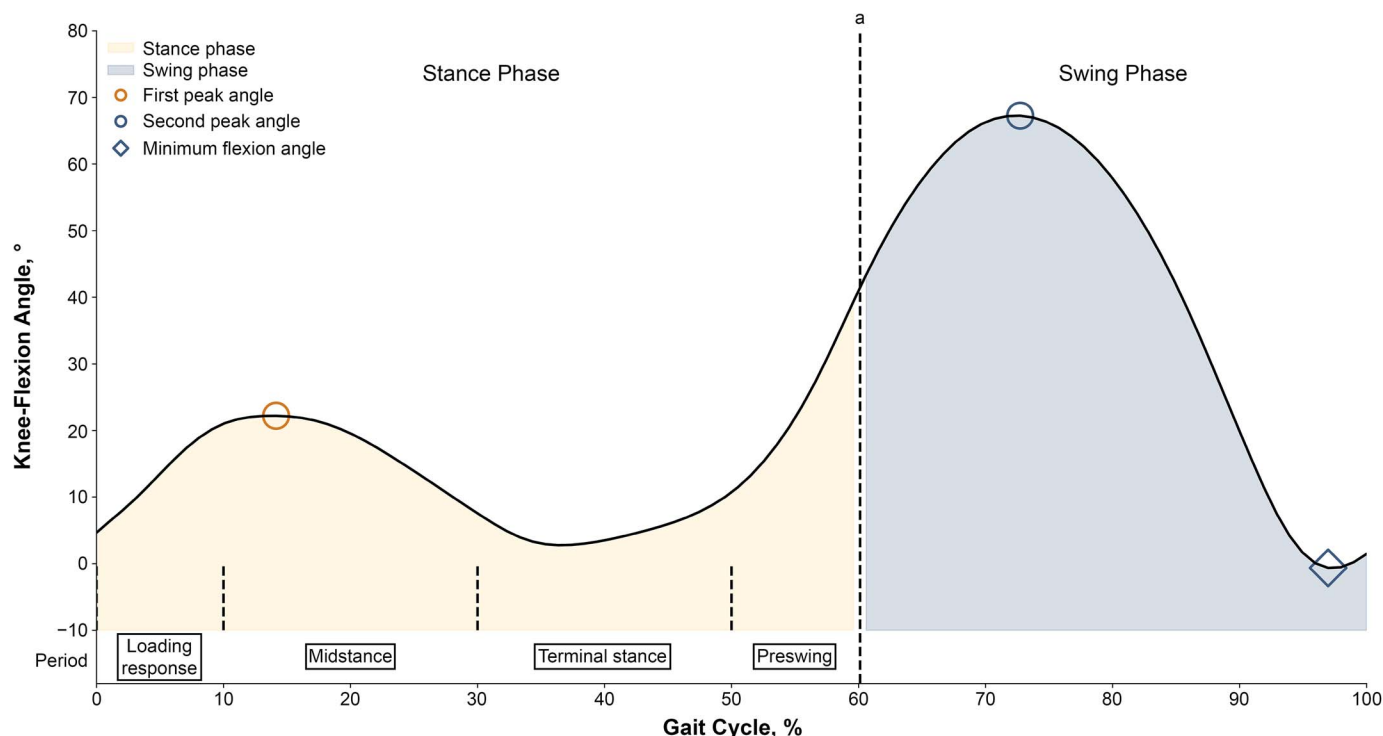


Figure 1. Knee sagittal-plane movement angle during the normalized gait cycle of healthy participants and commonly analyzed discrete points. ^a Point between the stance and swing phase.

Individuals were excluded if they had sustained multiligament injuries, substantial meniscus damage, or any other concurrent injuries that altered the standard weight-bearing protocol. We also excluded individuals who had a history of previous knee surgery or fracture in either limb, known neuropathies, active cancer, or inflammatory arthritis; had received an injection for pain in the 6 months before the study; or declined to participate. All participants followed a standardized baseline rehabilitation protocol (Appendix Table 1).

Volunteers matched by sex, weight, and height were recruited for a healthy control group and underwent a similar gait assessment protocol in a single testing session. The inclusion criteria were male or female volunteers aged 18 to 40 with no self-reported lower limb or back pain in the 6 months before the study. The exclusion criteria were a history of any lower limb or back fractures, surgeries, known neuropathies, active cancer, or inflammatory arthritis.

All participants provided written informed consent, and the study was approved by the Rambam Health Care Campus Helsinki Committee (0089-21-RMB).

Gait Assessment

Each participant was fitted with 7 IMUs (Xsens Awinda): 1 on each foot, 1 just below each knee, 1 on each mid thigh, and 1 on the pelvis. Data were collected, automatically processed, and filtered using proprietary software (MVN Analyze version 2023.0.0; Xsens) at a sampling rate of 100 Hz. The system captures acceleration, angular velocity, and magnetic field data; exports them to a computer; and processes them using biomechanical modeling algorithms and proprietary software to compute joint angles, spatiotemporal parameters, and segment orientations in 3 dimensions.²⁶

This system was shown to be reliable compared with traditional motion capture.²⁷

Before data collection, we performed a calibration process to ensure accurate sensor alignment. Next, participants were asked to walk across a 20-m corridor at 3 speeds: slow, normal, and fast. The following standardized oral instructions were used for slow, normal, and fast speeds, respectively: “Walk across the walkway at a slow speed,” “Walk across the walkway at your normal speed,” and “Walk as fast as possible.”²⁸ Each speed was repeated 3 times. The main discrete outcome was the knee sagittal-plane angles, which comprised the *first peak angle*, defined as the first peak of knee flexion during the stance phase; the *second peak angle*, defined as the peak of knee flexion occurring during the swing phase; and the *minimum flexion angle*, defined as the smallest knee-flexion angle reached during the gait cycle. The stance and swing phases were defined as 0% to 60% and 60% to 100% of the gait cycle, respectively. The stance phase was subdivided into *loading response* (0%–10%), *midstance* (10%–30%), *terminal stance* (30%–50%), and *preswing* (50%–60%; Figure 1). The secondary outcome was the minimum and maximum sagittal-plane angles of the hip and ankle. All outcomes were averaged and used for the final analysis. Participants were asked to bring their own walking shoes and wear the same shoes at each visit.

Statistical Analysis

Independent-samples *t* tests were used to compare continuous demographic variables, and χ^2 tests were used for demographic categorical variables. Descriptive statistics are presented as median (minimum, maximum). The

Table 1. Participant Characteristics

Characteristic	Participants With ACLR (n = 40)	Uninjured Participants (n = 17)	Test Statistic ^a	P Value
Sex, No. (%)			χ^2_1 (N = 57) = 1.02	.31
Male	33 (82.5)	12 (70.5)		
Female	7 (17.5)	5 (29.5)		
Age, mean $\pm$ SD, y	23.9 $\pm$ 5.9	28.4 $\pm$ 5.9	$t_{55} = -2.77$	.008
Height, mean $\pm$ SD, m	1.76 $\pm$ 0.10	1.70 $\pm$ 0.10	$t_{55} = -1.98$	.052
Mass, mean $\pm$ SD, kg	76.3 $\pm$ 13.4	68.3 $\pm$ 15.4	$t_{55} = -1.70$	.09
Injured leg, No. (%)			NA	N/A
Left	18 (45)	NA		
Right	22 (55)	NA		
Time between injury and reconstruction, median (minimum, maximum), d	191 (45, 698)	NA	NA	N/A
Graft type, No. (%)			NA	N/A
Hamstrings	26 (65)	N/A		
Bone-patellar tendon-bone	8 (20)	N/A		
Quadriceps	5 (12.5)	N/A		
Allograft	1 (0.02)	N/A		

Abbreviations: ACLR, anterior cruciate ligament reconstruction; NA, not available.

^a Test statistics are an independent-samples *t* test or χ^2 test of independence.

median and interquartile range are reported for walking speed for each walking condition grouped by time point. The Wilcoxon signed rank test with the exact method was used to compare the injured and contralateral limbs at each time point, and the Mann-Whitney *U* test was used to compare the injured limb and the left limb of the control group. The Friedman test was used to assess differences between walking speeds.

We used the Benjamini-Hochberg procedure to control for multiple comparisons and false discovery rates. We applied the Benjamini-Hochberg procedure separately for each aim, ensuring that the control of the false discovery rate was appropriate for each hypothesis and maintaining statistical power without the results of one aim affecting the other. The false discovery rate level was set at 0.1, balancing the need to detect true effects.²⁹

Linear mixed-effects models were used to assess longitudinal changes in the lower limbs. Each joint angle was the independent variable, the time point was added as a fixed effect, and the participants with ACLR were added as random effects. When differences were detected between time points, the Bonferroni post hoc test was used for pairwise comparisons between the different time points. The analyses were carried out using SPSS Statistics (version 29; IBM Corp) and Jamovi (version 2.4; The Jamovi Project).

Before performing the SPM analysis, we normalized all time-series data to 101 data points, representing 0% to 100% of the gait cycle. The data were interpolated to 101 evenly spaced points for each gait cycle using a linear interpolation function (`scipy.interpolate.interp1d`, SciPy version 1.15.0; SciPy). Discrete outcome measures were then extracted from these normalized gait cycles, with the knee sagittal-plane angles defined as the first peak angle, the second peak angle, and the minimum flexion angle. Next, 1-dimensional SPM (`spm1d` version 3.7; Python Software Foundation) was used to evaluate differences in the full kinematic wave during walking. The `spm1d` nonparametric paired-samples *t* test with 1000 permutations was used to

assess the limb differences. Mean $\pm$ 1 SD joint-angle trajectory across participants is reported for SPM. The α level for all tests was set at .05.

RESULTS

Participant Characteristics and Walking Speeds

Table 1 provides the descriptive data of the patients with ACLR and control participants. Knee kinematics are presented in Table 2. We observed no differences in sex, age, height, mass, and graft type between the participants who attended the 5-month follow-up and those who were lost to follow-up at 5 months post-ACLR (Appendix Table 2).

Figure 2 shows the difference in walking speeds at different time points. The results of the Friedman test showed differences between slow, normal, and fast walking speeds within each time point (presurgery: $\chi^2_2 = 80.0$, $P < .001$; 3 months: $\chi^2_2 = 52.0$, $P < .001$; and 5 months: $\chi^2_2 = 36.0$, $P < .001$). When comparing the same speed categories across time points, no differences were observed in slow ($\chi^2_2 = 1.88$, $P = .39$), normal ($\chi^2_2 = 4.35$, $P = .11$), or fast ($\chi^2_2 = 3.29$, $P = .19$) walking speeds, implying that speed was sustained over the 3 time points.

Hip and ankle kinematics are detailed in Appendix Tables 3 and 4, respectively.

Pre-ACLR Knee Kinematics

The discrete analysis showed that the injured limb exhibited greater knee flexion at the minimum flexion angle compared with the contralateral limb during normal ($M = 1.7^\circ$, $SE = 0.7^\circ$; $ES = 0.48$; $P = .006$) and fast ($M = 1.7^\circ$, $SE = 0.7^\circ$; $ES = 0.57$; $P = .008$) walking. The injured limb showed less knee flexion at the first peak angle ($M = -1.7^\circ$, $SE = 0.8^\circ$; $ES = -0.49$; $P = .006$).

The SPM analysis revealed that the injured knee was less flexed than the contralateral knee across multiple portions of the gait cycle (Table 3). During slow walking, differences were observed from 12% to 54% of the gait cycle

Table 2. Description of the Knee Sagittal-Plane Movements While Walking at 3 Time Points Comparing the Involved Limb With the Contralateral Limb and a Healthy Cohort^a

Walking Pace	Angle	Control (n = 17)			Pre-ACLR (n = 40)			3 mo Post-ACLR (n = 27)			5 mo Post-ACLR (n = 18)		
		Left Limb, °	Right Limb, °	P Value ^b	Involved Limb, °	Contralateral Limb, °	P Value ^b	Involved Limb, °	Contralateral Limb, °	P Value ^b	Involved Limb, °	Contralateral Limb, °	P Value ^b
Slow	Minimum	-3.8 (-10.2, 1.8)	-3.7 (-9.6, 1.7)	.24	-1.3 (-7.5, 14.6)	-3.5 (-10.8, 5.5)	.053 ^c	1.0 (-5.2, 7.9)	-3.5 (-7.9, 1.3)	<.001 ^{e,d}	-1.8 (-5.2, 6.9)	-2.7 (-6.5, 3.0)	.03 ^{e,d}
	First peak	5.4 (-6.2, 22.5)	7.1 (-3.6, 21.7)	.08	9.57 (-4.1, 21.6)	7.2 (-5.0, 21.5)	.07	11.2 (0.1, 22)	10.5 (-1.4, 23.5)	.63 ^c	10.0 (-0.6, 21.0)	11.5 (-3.0, 21.4)	.61
	Second peak	58.6 (48.1, 65.7)	59.5 (52.7, 66.3)	.23	58.7 (44.8, 71.7)	59.9 (53.7, 69.2)	.06	55.9 (38.5, 66.6)	59.7 (53.7, 70.0)	<.001 ^d	58.6 (48.3, 73.1)	59.5 (54.1, 70.9)	.37
Normal	Minimum	-4.3 (-9.4, -0.6)	-3.0 (-10.8, 1.1)	.03 ^b	-2.1 (-8.0, 14.0)	-3.9 (-11.7, 4.2)	.006 ^{e,d}	1.3 (-4.0, 9.0)	-4.2 (-8.2, 0.6)	<.001 ^{e,d}	-1.5 (-6.5, 6.6)	-3.2 (-7.4, 1.5)	.003 ^{e,d}
	First peak	13.7 (-2.6, 28.1)	13.4 (-2.6, 28.1)	.23	14.5 (-4.0, 25.3)	14.7 (-3.6, 27.9)	.62	13.1 (2.5, 23.8)	15.0 (-1.8, 24.6)	.26	13.0 (4.3, 21.6)	16.5 (0.7, 25.4)	.048 ^c
	Second peak	60.7 (56.4, 69.4)	61.4 (54.3, 71.0)	.26	63.3 (49.3, 71.0)	63.2 (58.5, 71.5)	.13	60.2 (37.1, 71.7)	63.1 (55.0, 73.9)	.001 ^c	62.6 (51.1, 73.8)	61.8 (57.7, 72.0)	.42
Fast	Minimum	-2.9 (-12.7, 1.7)	-1.6 (-10.3, 2.8)	.13	-1.5 (-7.0, 12.4)	-3.0 (-11.1, 4.5)	.008 ^c	1.8 (-3.1, 14.7)	-2.7 (-7.3, 3.6)	<.001 ^{e,d}	-0.6 (-5.7, 7.9)	-3.1 (-7.3, 2.2)	<.001 ^c
	First peak	18.6 (5.2, 33.2)	21.8 (-1.0, 30.9)	.09	18.8 (-4.4, 28.9)	20.3 (9.8, 29.9)	.006 ^c	17.1 (4.1, 29.8)	21.6 (12.2, 29.7)	<.001 ^c	17.0 (2.2, 25.4)	21.5 (-0.2, 31.1)	<.001 ^c
	Second peak	60.5 (55.2, 74.5)	59.8 (52.9, 70.4)	.78	63.4 (48.4, 71.4)	62.6 (53.7, 71.6)	.59	61.1 (51.7, 75.3)	61.9 (55.5, 71.2)	.005 ^c	62.4 (54.1, 73.7)	61.3 (57.7, 70.2)	.47

Abbreviation: ACLR, anterior cruciate ligament reconstruction.

^a Limb data are median (minimum, maximum).

^b P values represent between-limbs comparisons.

^c Indicates a difference between the involved and contralateral limbs of the ACLR group or right and left limbs of the control group at a single time point. Comparisons were made using the exact Wilcoxon signed-rank test.

^d Indicates a difference between the involved limb of the ACLR group at the chosen time and the left leg of the control group. Comparisons were made using the exact Mann-Whitney U test.

($t = 2.48$, $P = .001$; Figure 3). During normal walking, differences were observed from 24% to 48% of the cycle ($t = 2.75$, $P = .001$; Figure 4). Differences were noted during fast walking in 2 areas: 30% to 51% and 90% to 94% of the gait cycle ($t = 2.79$, $P = .002$ and $t = 2.79$, $P = .03$, respectively; Figure 5).

We observed that the injured knee of the patients with ACLR was more flexed at the minimum flexion angle compared with the knee of controls during slow ($M = 2.7^\circ$; $ES = 0.38$; $P = .02$) and normal ($M = 2.2^\circ$; $ES = 0.37$; $P = .03$) walking.

3 Months Post-ACLR Knee Kinematics

The discrete analysis indicated that, compared with the contralateral limb, the injured limb displayed greater knee flexion at the minimum flexion angle ($M = 4.3^\circ$, $SE = 0.6^\circ$; $ES = 0.96$; $P < .001$) and less knee flexion at the second peak angle ($M = -4.7^\circ$, $SE = 0.8^\circ$; $ES = -0.92$; $P < .001$) during slow walking. The injured limb had greater knee flexion at the minimum flexion angle ($M = 4.3^\circ$, $SE = 0.8^\circ$; $ES = 0.98$; $P < .001$) and less knee flexion at the first ($M = -5.2^\circ$, $SE = 0.9^\circ$; $ES = -0.86$; $P < .001$) and second ($M = -2.2^\circ$, $SE = 0.89^\circ$; $ES = -0.60$; $P = .005$) peak angles during fast walking.

The SPM analysis showed multiple periods in which the injured knee was less flexed across all walking speeds (Table 3). During slow walking, differences between limbs occurred across 3 clusters of the gait cycle: 0% to 3%, 19% to 61%, and 92% to 99% ($t = 2.63$, $P = .03$; $t = 2.62$, $P = .001$; and $t = 2.62$, $P = .001$, respectively; Figure 3). We observed differences in normal walking from 26% to 59% and 90% to 99% of the cycle ($t = 2.78$, $P = .001$ for both; Figure 4). During fast walking, differences were observed from 30% to 62% and 90% to 97% of the gait cycle ($t = 2.86$, $P = .001$ and $t = 2.86$, $P = .008$, respectively; Figure 5).

Compared with the control knee, the injured knee remained more flexed at the minimum flexion angle at slow ($M = 5.3^\circ$; $ES = 0.78$; $P < .001$), normal ($M = 5.5^\circ$; $ES = 0.77$; $P < .001$), and fast ($M = 4.9^\circ$; $ES = 0.71$; $P < .001$) walking speeds. The knee was less flexed at the first peak angle during slow walking ($M = 4.6^\circ$; $ES = 0.39$; $P = .03$).

5 Months Post-ACLR Knee Kinematics

Compared with the contralateral knee, the discrete analysis showed that the injured knee was more flexed at the minimum flexion angles during slow ($M = 1.5^\circ$, $SE = 0.6^\circ$; $ES = 0.59$; $P = .03$) and normal ($M = 2.1^\circ$, $SE = 0.7^\circ$; $ES = 0.76$; $P = .003$) walking speeds. The injured knee was less flexed at the first peak angle during normal walking speed ($M = -1.9^\circ$, $SE = 1.0^\circ$; $ES = -0.53$; $P = .048$), but this finding was not different after correction.

Similarly, during fast walking, the injured knee was more flexed at the minimum flexion angle ($M = 2.3^\circ$, $SE = 0.6^\circ$; $ES = 0.86$; $P < .001$) and less flexed at the first peak angle ($M = -5.1^\circ$, $SE = 1.0^\circ$; $ES = -0.83$; $P < .001$).

Figures 4 and 5 show that differences were limited to 93% to 95% of the gait cycle during normal walking ($t = 2.86$, $P = .003$) and 38% to 61% during fast walking ($t = 2.83$, $P = .01$). The injured knee remained less flexed during the loading phase during fast walking compared with

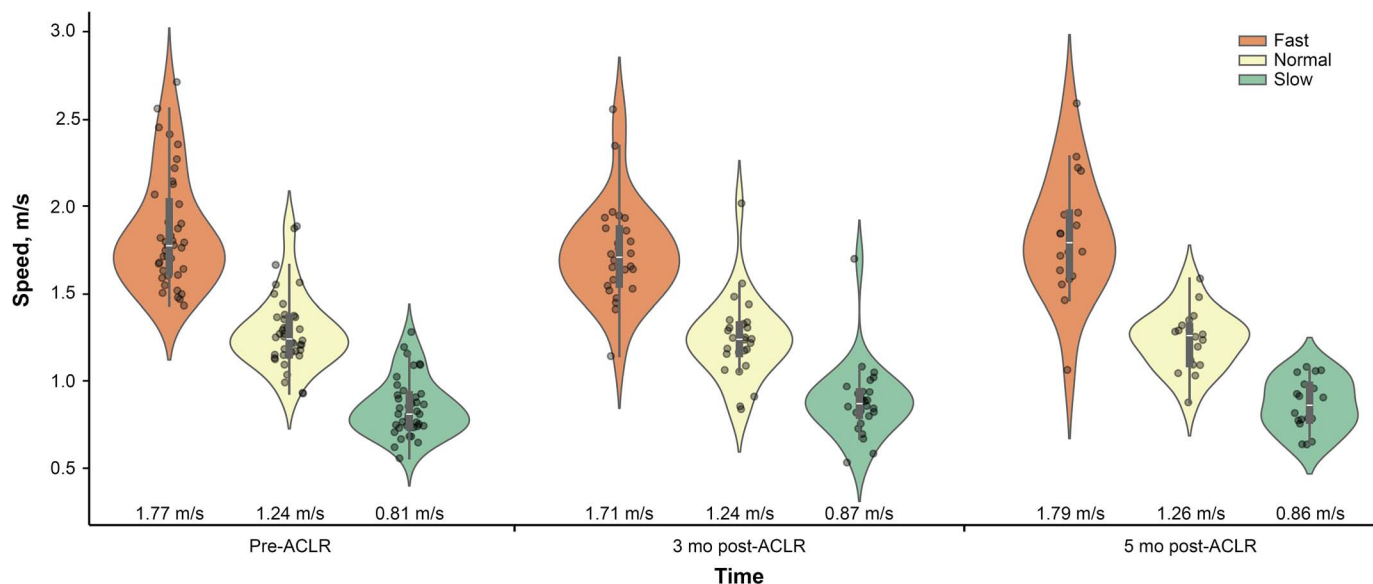


Figure 2. Median walking speed for each walking condition grouped by time point. Violin plots represent the probability density of data at different speed values, with the width of the violin indicating the frequency of observations at each speed. Black lines within the violins indicate the median [interquartile range] speed of each group. Individual data points (gray circles) overlaid on the violin plots represent the interparticipant differences in speed values.

the contralateral knee, indicating that some asymmetries persisted, mostly at faster walking speeds. Compared with the control knee, the injured knee was more flexed at the minimum angle at slow ($M = 3.2^\circ$; $ES = 0.57$; $P = .003$) and normal ($M = 3.3^\circ$; $ES = 0.54$; $P = .005$) walking speeds.

As seen in Table 4, changes were observed in knee kinematics from pre-ACLR to 3 months post-ACLR. At slow walking speed, knee flexion was increased for the minimum angle ($P = .006$). Similar patterns were observed at normal ($P < .001$) and fast ($P = .006$) walking speeds.

Associated Hip and Ankle Adaptations

After correction for multiple comparisons, differences were observed at pre-ACLR in the ankle of the injured limb that demonstrated increased plantarflexion (slow speed, $P = .003$) compared with the controls. No differences were noted between the injured and contralateral limbs. By 3 months post-ACLR, the hip showed less flexion across all walking speeds (normal and slow, $P = .001$ and fast, $P = .008$) and less extension at normal walking speed ($P = .004$) in the injured than contralateral limb. The

Table 3. Results of a Nonparametric Paired t Test Using 1-Dimensional Statistical Parametric Mapping^a Analysis for Differences Between the Injured and Contralateral Limbs During Walking at 3 Different Speeds by Joint and Time

Walking Pace	Joint	Pre-ACLR (n = 40)			3 mo Post-ACLR (n = 27)			5 mo Post-ACLR (n = 18)		
		Clusters, % Cycle	t^b	P Value	Clusters, % Cycle	t^b	P Value	Clusters, % Cycle	t^b	P Value
Slow	Knee	12, 54	2.48	.001 ^c	0, 3 19, 61 92, 99	2.63	.03 ^c .001 ^c .001 ^c	d	d	d
	Hip	12, 37	2.56	.01 ^c	24, 50	2.68	.004 ^c	d	d	d
	Ankle	20, 43	2.69	.001 ^c	19, 48 67, 91	2.89	.001 ^c .02 ^c	d	d	d
Normal	Knee	24, 48	2.75	.001 ^c	26, 59 90, 99	2.78	.001 ^c .001 ^c	93, 95	2.86	.003
	Hip	d	d	d	34, 47	2.71	.01 ^c	d	d	d
	Ankle	29, 36	2.79	.03 ^c	23, 44 85, 88	2.88	.001 ^c .045 ^c	34, 37	2.98	.04
Fast	Knee	30, 51 90, 94	2.79	.002 ^c .03 ^c	30, 62 90, 97	2.86	.001 ^c .008 ^c	38, 61	2.83	.01
	Hip	d	d	d	34, 57	2.75	.006 ^c	d	d	d
	Ankle	d	d	d	27, 41	2.85	.001 ^c	33, 39	3.03	.02

Abbreviation: ACLR, anterior cruciate ligament reconstruction.

^a A permutation-based approach with 1000 iterations was used for cluster level.

^b A positive t value indicates a larger angle for the injured limb.

^c Indicates difference ($P < .05$).

^d Indicates no difference between limbs.

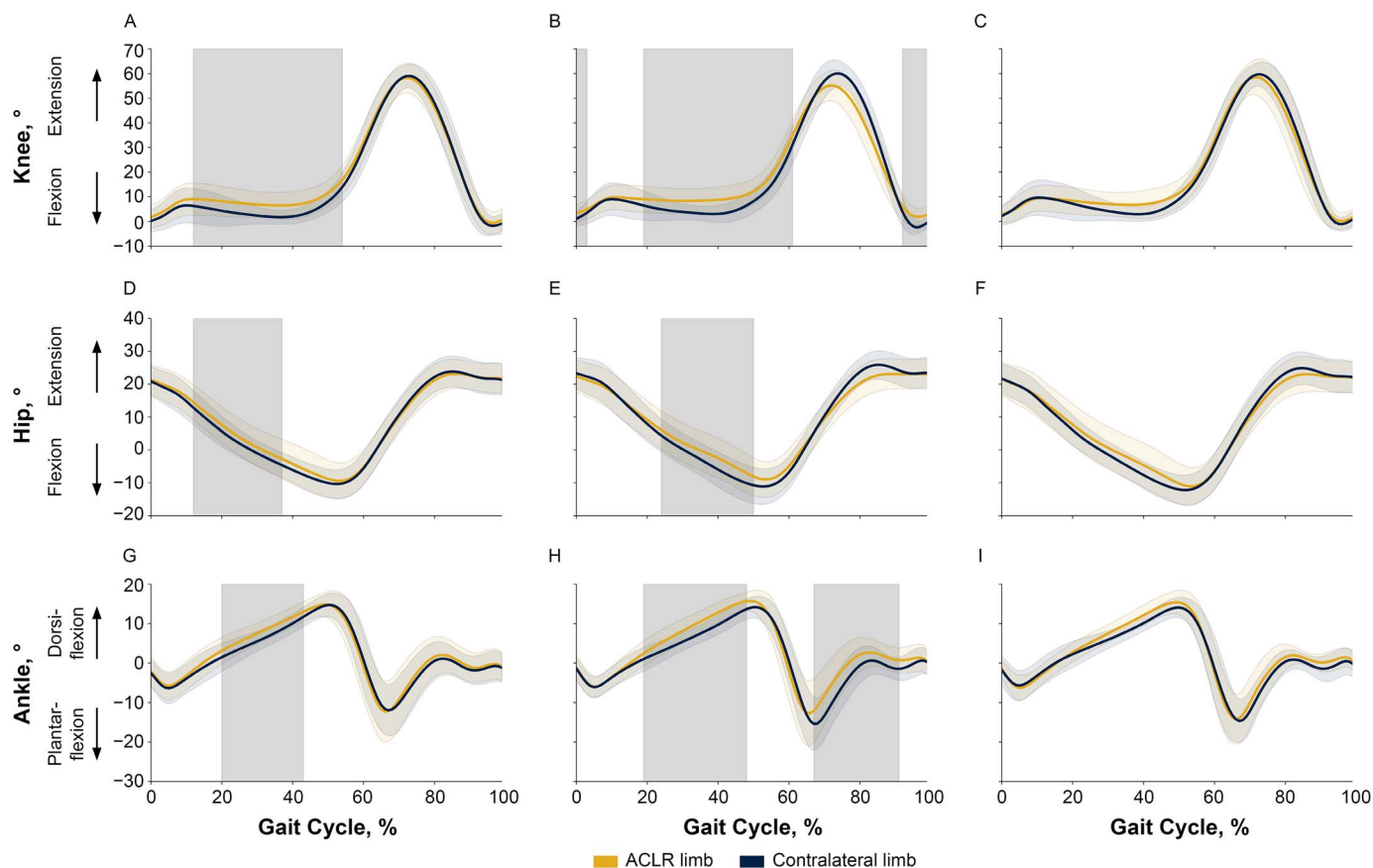


Figure 3. One-dimensional statistical parametric mapping of the gait cycle for the anterior cruciate ligament reconstruction (ACLR) and contralateral limbs during slow walking at pre-ACLR and 3 and 5 months post-ACLR, respectively, for the, A–C, knee, D–F, hip, and G–I, ankle. Solid lines indicate the mean joint-angle trajectory across participants, with shaded areas indicating ± 1 SD. Shaded gray boxes indicate differences.

ankle of the injured limb exhibited increased plantarflexion (slow speed, $P = .002$).

By 5 months post-ACLR, no differences were found in the hip joints between limbs and between the injured and control limbs. The ankle of the injured limb showed less dorsiflexion and increased plantarflexion at slow walking speed ($P < .05$) compared with controls, but those findings were not different after correction for multiple comparisons. The results of the SPM analysis are available in Table 3, and the discrete analysis is provided in Appendix Tables 3 to 6.

DISCUSSION

We assessed sagittal-plane kinematics during gait pre-ACLR and post-ACLR using IMUs in a nonlaboratory setting. Our primary findings indicated progressive improvement in knee sagittal-plane kinematics symmetry post-ACLR, particularly at 3 months post-ACLR, with compensatory adaptations observed in hip and ankle kinematics. Few researchers have examined pre-ACLR and early post-ACLR gait kinematics or the effect of walking speed on gait asymmetries or have used SPM in this population.^{5,11,12,19,30}

Pre-ACLR Knee Kinematics

The injured knee exhibited a more flexed position during most of the stance phase across all speeds and just before the loading phase at fast speed. These results are consistent

with those of Gao and Zheng, who observed that patients who were <1 year since their ACL injury presented with a less extended knee during the stance phase.³¹ The SPM analysis revealed that these asymmetries occurred predominantly during midstance and terminal stance, highlighting how weight-bearing activities specifically challenge movement symmetry even before surgery. Increased knee flexion during stance may represent a protective mechanism to reduce tibial translation and minimize ACL strain in the absence of a functional ACL.³² The persistence of this pattern across different walking speeds suggests it is a robust compensatory strategy rather than a speed-dependent adaptation.

Three Months Post-ACLR Knee Kinematics

The discrete analysis found that, at 3 months post-ACLR, participants demonstrated less knee flexion at the second peak during slow and normal walking speeds in their injured than contralateral limb. During fast walking, differences between limbs extended to include less flexion at the first peak angle in the injured knee.

Whereas Williams et al reported smaller first peak knee-flexion angle in the involved than uninvolved limb during self-selected walking speeds ($17.7^\circ \pm 6.7^\circ$ versus $20.9^\circ \pm 4.7^\circ$), we only observed this difference during fast walking.³³ This discrepancy may be explained by walking speed differences, as the self-selected speeds of their participants were faster than those of ours, falling between our normal

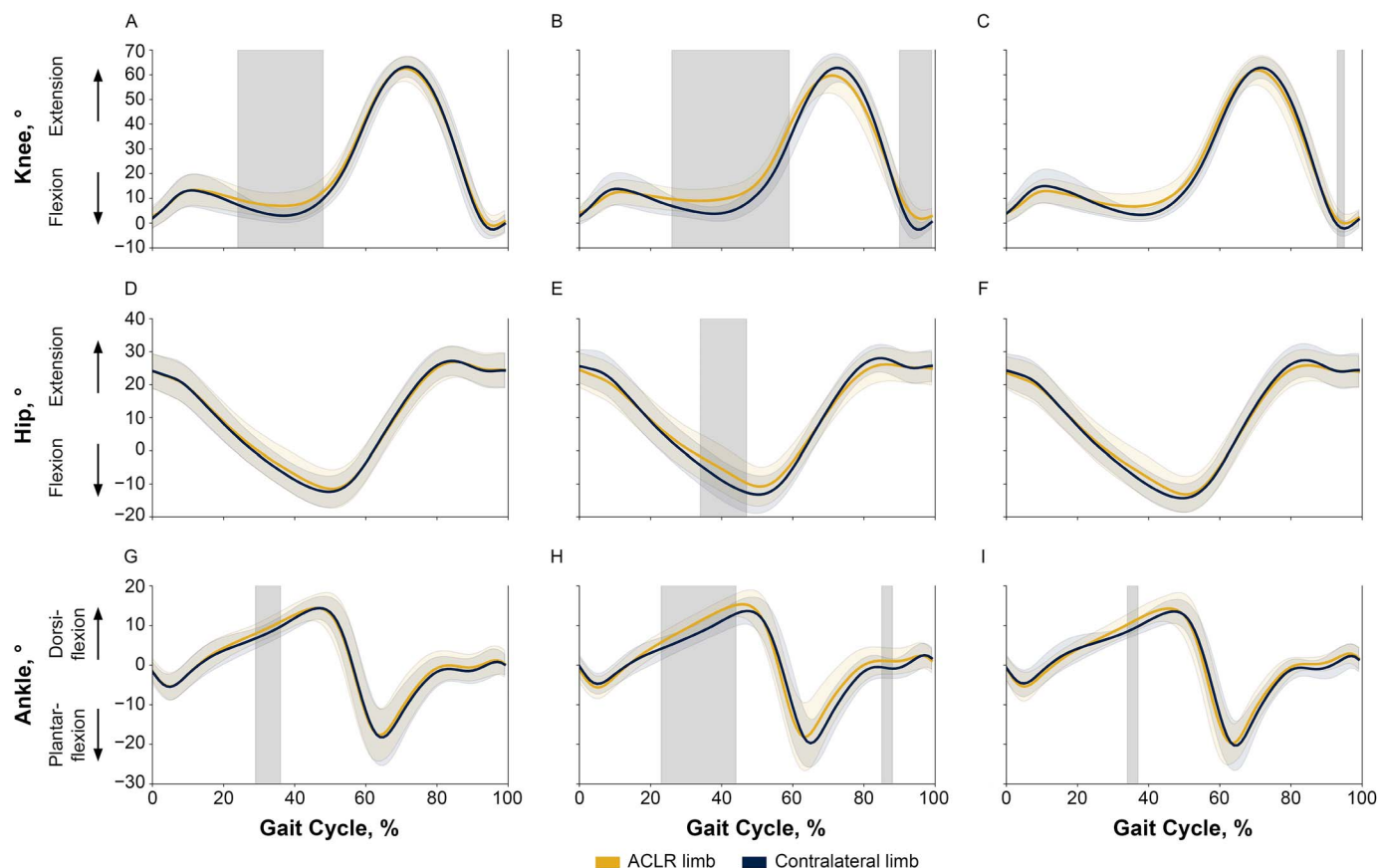


Figure 4. One-dimensional statistical parametric mapping of the gait cycle for anterior cruciate ligament reconstruction (ACLR) and contralateral limbs during normal walking speed at pre-ACLR and 3 and 5 months post-ACLR, respectively, for the, A–C, knee, D–F, hip, and G–I, ankle. Solid lines indicate the mean joint-angle trajectory across participants, with shaded areas indicating ± 1 SD. Shaded gray boxes indicate differences.

and fast conditions. Furthermore, Ferber et al found no difference in first-peak knee-flexion angle at a self-selected walking speed but did not report on the actual walking speed of the participants (control = $16.1^\circ \pm 2.1^\circ$; ACLR = $19.8^\circ \pm 5.4^\circ$).³⁴

The SPM analysis indicated that the injured knee was more flexed during most of stance and the beginning of the swing phase across all walking speeds but had less peak flexion during the swing phase. Neal et al used SPM to assess knee kinematics 3 months post-ACLR and reported increased knee flexion between 31% to 51% of the gait cycle, consistent with our findings.¹⁸ On the contrary, they found decreased knee flexion between 6.5% and 20% of the gait cycle. Notably, the self-selected speed in the study by Neal et al was closer to our participants' fast walking speed, partially explaining this discrepancy.¹⁸

These findings reinforce the speed-dependent nature of kinematic asymmetries post-ACLR, suggesting that movement deficits may be less apparent at slower walking speeds but become more pronounced as task demands increase. Furthermore, the late stance and terminal swing asymmetries might highlight challenges in preparing for the next gait cycle.

Five Months Post-ACLR Knee Kinematics

By 5 months post-ACLR, the injured limb improved, but asymmetries persisted in terminal stance to preswing at fast

walking speed. This speed-dependent pattern demonstrates how increasing task demands can reveal persistent movement asymmetries that might not be apparent in less challenging walking.

Discrete analysis revealed that the knee was more flexed at the minimum flexion angle and less flexed at the first peak at normal and fast speeds compared with the contralateral limb. These results differ from those of Georgoulis et al, who assessed normal walking speed (1.28 m/s) and found no differences at any discrete point of the gait cycle at a walking speed similar to that in our study (1.26 m/s).³⁵ One explanation is that Georgoulis et al compared 2 separate groups (different participants for the pre-ACLR and post-ACLR groups), whereas we performed within-subjects comparisons.³⁵

Neal et al assessed participants 6 months post-ACLR and found differences between 5% to 18% and 36% to 48% of the gait cycle.¹⁸ Similar to the 3-month comparison, the self-selected speed was faster in their study than ours. The differences in sample size, movement-capture systems, and slightly different time points post-ACLR can further explain the different results.

Although improvements during the first 5 months post-ACLR are evident, knee kinematics do not fully return to preinjury levels even 1 year after surgery. Researchers have shown that ACLR limbs have less peak knee flexion compared with control limbs, which could contribute to less-than-optimal recovery.² Furthermore, the persistence of

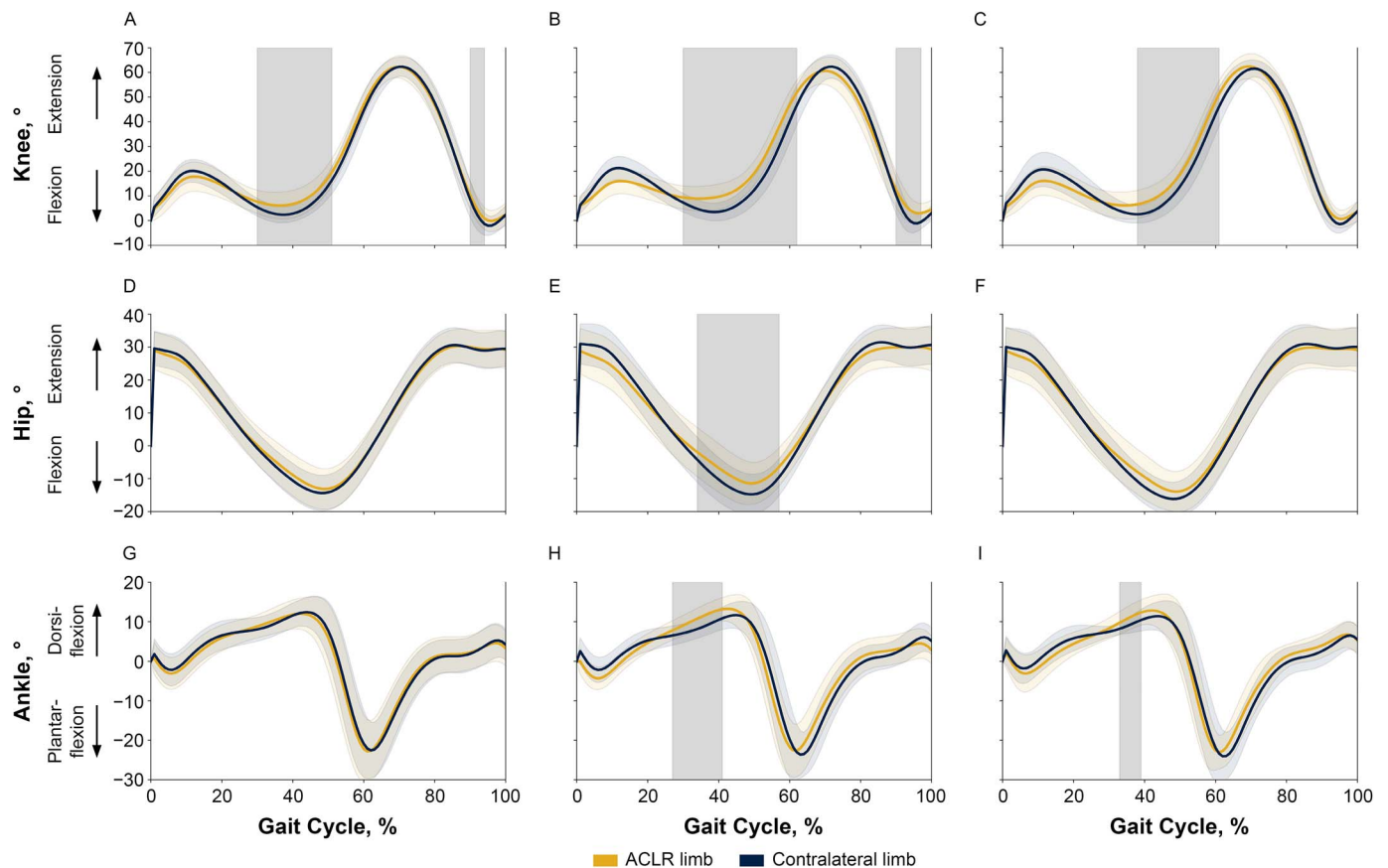


Figure 5. One-dimensional statistical parametric mapping of the gait cycle for anterior cruciate ligament reconstruction (ACLR) and contralateral limbs during fast walking speed at pre-ACLR and 3 and 5 months post-ACLR, respectively, for the, A–C, knee, D–F, hip, and G–I, ankle. Solid lines indicate the mean joint-angle trajectory across participants, with shaded areas indicating ± 1 SD. Shaded gray boxes indicate differences.

asymmetries primarily during midstance at faster speeds might suggest that the neuromuscular system still struggles with higher mechanical demands, such as excessive tibial translation.³⁶

Associated Hip and Ankle Adaptations

As a secondary finding, we observed several adaptations in hip and ankle kinematics, possibly as a compensation strategy for the changes in the knee joint. Pre-ACLR, hip asymmetries were limited to midstance to terminal stance during slow walking. Ankle asymmetries during this period

were evident in midstance to terminal stance when walking at slow and normal speeds. At 3 months post-ACLR, hip asymmetries persisted during midstance, terminal stance, or preswing across all walking speeds. Similarly, ankle asymmetries were observed in midstance and terminal stance across all walking speeds. By 5 months post-ACLR, hip asymmetries had resolved across all speeds, but ankle asymmetries persisted during terminal stance at normal and fast walking.

The resolution of hip asymmetries by 5 months compared with the persistence of ankle asymmetries suggests a joint-specific recovery pattern. These findings suggest

Table 4. Description of Changes in the Knee Sagittal-Plane Movements Between the Different Time Points Before and After ACLR

Walking Pace	Angle	Pre-ACLR to 3 mo Post-ACLR		3 mo to 5 mo Post-ACLR		Pre-ACLR to 5 mo Post-ACLR	
		Difference, Mean $\pm$ SD, °	P Value	Difference, Mean $\pm$ SD, °	P Value	Difference, Mean $\pm$ SD, °	P Value
Slow	Minimum	-2.5 ± 0.7	.006 ^a	2.2 ± 0.8	.045 ^a	-0.3 ± 0.9	>.99
	First peak	-1.0 ± 0.9	.77	1.0 ± 0.7	.41	0.1 ± 1.0	>.99
	Second peak	3.0 ± 1.1	.04 ^a	-2.9 ± 1.4	.14	0.1 ± 1.2	>.99
Normal	Minimum	-2.9 ± 0.7	<.001 ^a	2.2 ± 0.8	.55	-0.7 ± 0.9	>.99
	First peak	0.1 ± 0.9	>.99	0.3 ± 0.5	>.99	0.4 ± 0.9	>.99
	Second peak	3.0 ± 1.2	.07	-1.8 ± 1.5	.69	1.1 ± 0.9	.73
Fast	Minimum	-3.1 ± 0.9	.006 ^a	2.7 ± 1.0	.042 ^a	-0.4 ± 1.0	>.99
	First peak	1.2 ± 1.0	.72	0.45 ± 1.0	>.99	1.6 ± 1.0	.43
	Second peak	1.8 ± 0.9	.16	-1.6 ± 0.9	.31	0.1 ± 0.7	>.99

Abbreviation: ACLR, anterior cruciate ligament reconstruction.

^a Indicates difference between time points.

compensations still exist, particularly under greater demands, which can lead to greater joint loading.³⁷

Clinical Implications

The comparison of our study with others highlights the need for transparent, reproducible methodologies. Given that some researchers did not report walking speeds and others did not describe the exact oral cue given to their participants, a heterogeneous comparison between studies is hard to make.^{34,38}

Our findings using IMU-based gait analysis aligned with previous evidence supporting the clinical utility of wearable sensors in assessing knee pathologies. In a recent study, Ghaffari et al demonstrated that IMU-derived gait variables can be used to discriminate between individuals with and those without knee OA.³⁸ In addition, IMU measurements have shown associations with patient-reported outcomes and pain levels, suggesting their potential value as objective markers of functional recovery.^{39,40} Whereas researchers have primarily focused on primary knee OA, we extended these findings to the ACLR population, demonstrating that IMU-based gait analysis can capture subtle biomechanical changes and potentially early identification of patients at risk for developing PTOA.

Our findings have important implications for post-ACLR rehabilitation protocol design and implementation. The nonlinear recovery patterns across walking speeds align with those of previous studies in which increased asymmetries at faster speeds were shown.^{11,30,41} Our secondary finding of compensatory mechanisms, particularly evident at higher speeds, may lead to excessive joint loading that could increase the risk of early degenerative changes in both the hip and ankle joints.^{41–44} Furthermore, different walking speeds can reveal important insights: Slower walking speeds may highlight joint patterns similar to those seen in OA, showing how patients with ACLR adapt their gait.⁴⁵ In addition, faster walking speeds place greater demands on joint moments, muscle activation, and range of motion.³⁰

Limitations

We acknowledge some limitations in our study. Although our sample size was determined by a previous trial rather than an a priori power calculation, sensitivity power analyses demonstrated our study had sufficient power (80%) to detect ESs of ≥ 0.45 , aligning with the effects we observed (range, 0.48–0.98). We used a single-center design with a limited follow-up period. However, the purpose of our study was to describe the kinematic changes during the pre-ACLR and early to middle stages of ACLR rehabilitation. We did not account for different graft types and sex differences, possibly introducing bias in our results. Furthermore, the participants available for this study had varied times between their injury and ACLR, which could have influenced their biomechanics and ACLR outcomes. Next, multiple comparisons of knee-, hip-, and ankle-joint kinematics might have increased type I errors. However, we predefined the knee-joint kinematic analysis as the primary outcome of this study and adjusted the results accordingly. In addition, the high dropout rate may have introduced attrition bias, potentially linked to participants missing predefined check-up meetings with their

surgeon rather than the study itself. Lastly, although we observed no differences in characteristics between the control and ACLR cohorts, the control participants were older and shorter and weighed less, which could have influenced their gait.

CONCLUSIONS

In our study, we highlighted several key findings regarding gait recovery in patients with ACLR. We observed improved symmetry in gait post-ACLR compared with the contralateral limb and controls. However, these improvements were inconsistent across different walking speeds, suggesting that varying walking speeds may challenge patients differently. Although hip and ankle adaptations were noted, the primary focus on knee kinematics revealed persistent asymmetries at faster walking speeds. These findings underscore the importance of integrating varied speed regimens in rehabilitation protocols to facilitate comprehensive recovery of knee function.

ACKNOWLEDGMENTS

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Appendix

Appendix Table 1. Baseline Rehabilitation Protocol

First week postoperatively
• Static quadriceps exercises in straightening. • Walking with crutches or walker while applying partial weight on the operated leg. • Wearing elastic socks and exercising calf muscles to avoid deep vein thrombosis.
7–14 d postoperatively
• Walking while fully stepping on the leg (can use crutches). • Exercising to improve range of movement. • After 2–3 wk, maximal recommended range of movement is 0°–90°.
Weeks 2–3
• Wall slides and heel slides exercises. • Static vastus medialis obliquus exercises sitting down. • Using rowing machine with minimal resistance and medium range of movement.
Week 4
• Bending exercises 0°–100° (avoid overstraightening). • Static bicycles with low resistance. • Up and down exercises with a low step. • Squat exercises up to 45°–60°.
Week 8
• Step machine. • Squat exercises.
Week 10
• Trampoline.
Week 12
• Achieving a range of movement of 0°–130° is recommended. • Advancing in strengthening quadriceps and hamstrings.
3–6 mo
• Increasing gradual activity. • Easy running starting in the 5th month. • Changing directions, accelerations, deceleration, running in figure of 8. • Preparing for a specific sport.
After 6 mo
• If everything is normal and the patient is confident to return to the specific sport, gradually exercise the specific sport.
7–9 mo
• Returning to the specific sport.

Appendix Table 2. Characteristics of Participants at 5-mo Follow-Up and Participants Lost to Follow-Up at 5 mo

Characteristic	Participants at 5 mo Post-ACLR (n = 18)	Participants Lost to Follow-Up at 5 mo (n = 22)	Test Statistic ^a	P Value
Sex, No. (%)			χ^2_1 (N = 40) = 2.39	.12
Male	13 (72.2)	20 (90.9)		
Female	5 (27.8)	2 (9.1)		
Age, mean $\pm$ SD, y	24.1 $\pm$ 6.6	23.6 $\pm$ 4.6	$t_{38} = -0.25$	.80
Height, mean $\pm$ SD, m	1.76 $\pm$ 0.10	1.76 $\pm$ 0.10	$t_{38} = -0.09$	.93
Mass, mean $\pm$ SD, kg	74.1 $\pm$ 12.6	75.7 $\pm$ 13.0	$t_{38} = 0.40$	.68
Body mass index, mean $\pm$ SD, kg/m ²	23.9 $\pm$ 3.3	24.6 $\pm$ 4.5	$t_{38} = 0.53$	.60
Injured limb, No. (%)			χ^2_1 (N = 40) = 0.60	.44
Left	6 (33.3)	10 (45.5)		
Right	12 (66.7)	12 (54.5)		
Graft type, No. (%)			χ^2_3 (N = 40) = 1.94	.59
Hamstrings	11 (61.1)	15 (68.2)		
Bone-patellar tendon-bone	3 (16.7)	5 (22.7)		
Quadriceps	3 (16.7)	2 (9.1)		
Allograft	1 (5.6)	0 (0)		

^a Independent-samples *t* test or χ^2 test of independence.**Appendix Table 3. Description of Changes in the Hip Sagittal-Plane Movements Between the Different Time Points Pre-ACLR and Post-ACLR**

Walking Pace	Angle	Pre-ACLR to 3 mo Post-ACLR		3 mo to 5 mo Post-ACLR		Pre-ACLR to 5 mo Post-ACLR	
		Difference, Mean $\pm$ SD, °	P Value	Difference, Mean $\pm$ SD, °	P Value	Difference, Mean $\pm$ SD, °	P Value
Slow	Minimum	0.7 $\pm$ 1.2	>.99	1.5 $\pm$ 0.6	.12	2.2 $\pm$ 1.2	.26
	Maximum	-0.4 $\pm$ 0.4	.83	0.4 $\pm$ 0.5	>.99	-0.1 $\pm$ 0.5	>.99
Normal	Minimum	0.1 $\pm$ 1.4	>.99	1.6 $\pm$ 1.1	.50	1.7 $\pm$ 1.0	.27
	Maximum	-0.1 $\pm$ 0.3	>.99	0.7 $\pm$ 0.4	.36	0.6 $\pm$ 0.4	.43
Fast	Minimum	-1.0 $\pm$ 1.5	>.99	1.1 $\pm$ 1.1	.96	0.1 $\pm$ 1.2	>.99
	Maximum	-0.1 $\pm$ 0.4	>.99	-0.4 $\pm$ 0.5	>.99	-0.4 $\pm$ 0.6	>.99

Abbreviation: ACLR, anterior cruciate ligament reconstruction.

Appendix Table 4. Description of Changes in the Ankle Sagittal-Plane Movements Between Time Points Pre-ACLR and Post-ACLR

Walking Pace	Angle	Pre-ACLR to 3 mo Post-ACLR		3 mo to 5 mo Post-ACLR		Pre-ACLR to 5 mo Post-ACLR	
		Difference, Mean $\pm$ SD, °	P Value	Difference, Mean $\pm$ SD, °	P Value	Difference, Mean $\pm$ SD, °	P Value
Slow	Minimum	-1.0 $\pm$ 0.6	.40	1.8 $\pm$ 0.8	.12	0.8 $\pm$ 0.7	.83
	Maximum	-0.2 $\pm$ 0.8	>.99	-0.3 $\pm$ 1.0	>.99	-0.6 $\pm$ 1.0	>.99
Normal	Minimum	-1.0 $\pm$ 0.7	.51	1.8 $\pm$ 0.8	.10	0.8 $\pm$ 0.7	.91
	Maximum	0.1 $\pm$ 0.8	>.99	-0.3 $\pm$ 0.9	>.99	-0.1 $\pm$ 0.8	>.99
Fast	Minimum	-2.0 $\pm$ 0.8	.06	1.9 $\pm$ 1.1	.30	-0.1 $\pm$ 1.0	>.99
	Maximum	-0.4 $\pm$.08	>.99	-0.6 $\pm$ 1.4	>.99	-1.1 $\pm$ 1.3	>.99

Abbreviation: ACLR, anterior cruciate ligament reconstruction.

Appendix Table 5. Hip Sagittal-Plane Movements While Walking at 3 Different Speeds at 3 Time Points Comparing the Involved Limb to the Contralateral Limb and a Healthy Cohort^a

Walking Pace	Angle	Control (n = 17)			Pre-ACLR (n = 40)			3 mo Post-ACLR (n = 27)			5 mo Post-ACLR (n = 18)		
		Left Limb, °	Right Limb, °	P Value	Involved Limb, °	Contralateral Limb, °	P Value	Involved Limb, °	Contralateral Limb, °	P Value	Involved Limb, °	Contralateral Limb, °	P Value
Slow	Minimum	-11.2 (-27.6, 20.3)	-11.7 (-24.0, -5.1)	.07	-10.2 (-20.2, 4.1)	-10.3 (-17.9, 0.6)	.28	-10.6 (-17.8, 4.2)	-10.7 (-25.3, -4.3)	.052	-11.8 (-21.0, 4.8)	-13.3 (-23.8, -5.9)	.37
	Maximum	25.4 (6.5, 31.4)	26.7 (13.9, 32.1)	.03 ^b	24.6 (15.1, 34.9)	24.6 (16.0, 37.1)	.28	24.3 (15.6, 32.1)	25.0 (16.5, 37.7)	<.001 ^b	23.1 (15.4, 33.4)	23.3 (18.8, 36.6)	.13
Normal	Minimum	-13.2 (-26.3, -8.1)	-12.9 (-24.8, -7.2)	.08	-11.8 (-23.2, 4.85)	-12 (-24.6, -3.5)	.20	-11.9 (-21.3, 2.2)	-11.3 (-27.4, -5.2)	.03 ^b	-14.5 (-21.0, 2.7)	-13.9 (-26.7, 14.3)	.97
	Maximum	28.6 (12.6, 34.8)	29.9 (17.0, 35.2)	.07	26.9 (18.6, 38.8)	27.4 (19.3, 41.5)	.30	27.6 (18.1, 34.9)	27.9 (20.0, 39.7)	<.001 ^b	26 (17.9, 35.2)	26.5 (22.4, 38.4)	.14
Fast	Minimum	-14.0 (-28.5, -9.3)	-13.6 (-25.5, -9.4)	.10	-14.2 (-26.9, 4.6)	-13.4 (-28.4, -2.3)	.17	-13 (-23.7, 2.2)	-13.6 (-32.1, -7.4)	.004 ^b	-15.4 (-22.6, 3.6)	-15.8 (-29.7, -7.2)	.054
	Maximum	34.3 (15.1, 42.4)	35.4 (16.7, 43.6)	.02 ^b	30.7 (18.5, 43.8)	30.2 (20.9, 44.6)	.38	30.5 (18.9, 45.5)	31.4 (22.6, 44.1)	.008 ^b	31.5 (15.1, 46.2)	32.6 (18.8, 46.2)	.13

Abbreviation: ACLR, anterior cruciate ligament reconstruction.

^a Limb data are median (minimum, maximum).

^b Indicates a difference between the involved and contralateral limbs of the ACLR group or right and left limbs of the control group at a single time point. The comparisons were made using the exact Wilcoxon signed-rank test.

Appendix Table 6. Ankle Sagittal-Plane Movements While Walking at 3 Time Points in Comparing the Involved Limb to the Contralateral Limb and a Healthy Cohort^a

Walking Pace	Angle	Control (n = 17)			Pre-ACLR (n = 40)			3 mo Post-ACLR (n = 27)			5 mo Post-ACLR (n = 18)		
		Left Limb, °	Right Limb, °	P Value	Involved Limb, °	Contralateral Limb, °	P Value	Involved Limb, °	Contralateral Limb, °	P Value	Involved Limb, °	Contralateral Limb, °	P Value
Slow	Minimum	-20.3 (-44.7, -9.8)	-17.4 (-36.7, -7.5)	.005 ^a	-13.4 (-30.0, 1.0)	-13.9 (-27.1, -0.2)	.86 ^b	-16.1 (-27.0, -2.15)	-18.6 (-26.3, -5.4)	.046 ^{ab}	-17.1 (-28.7, -7.6)	-17.2 (-27.6, -7.2)	.50 ^a
	Maximum	14.0 (7.7, 17.9)	14.5 (4.5, 20.2)	.43	15.5 (7.3, 21.8)	14.7 (7.4, 21.8)	.59 ^b	16.5 (11.6, 19.9)	14.8 (8.8, 19.0)	.02 ^{b,c}	15.5 (9.7, 21)	14.3 (11.0, 19.6)	.11 ^b
Normal	Minimum	-22.0 (-44.0, -15.2)	-21.2 (-41.1, -12.4)	.15	-18.7 (-33.8, -4.9)	-18.2 (-36.2, 16.9)	.92	-20.5 (-38.9, -4.0)	-22.2 (-32.8, -9.2)	.39	-20.1 (-35.1, -14.2)	-22.1 (-30.1, -11.0)	.67
	Maximum	13.7 (10.6, 20.6)	14.8 (7.2, 21.1)	.55	16.1 (7.2, 21)	14.9 (6.6, 20.6)	.57	16.2 (10.7, 21.2)	14.4 (7.5, 18.8)	.044 ^b	15.4 (8.6, 20.6)	14.5 (8.1, 19.2)	.30
Fast	Minimum	-27.6 (-46.7, -17.2)	-24.8 (-45.2, -17.8)	.26	-24.1 (-46.5, -7.3)	-23.4 (-41.8, -13)	.97	-24.7 (-36.9, -8.5)	-28.7 (-33.7, -14.7)	.10	-24.8 (-30.1, -18.6)	-27.6 (-36.1, -18.9)	.28
	Maximum	12.9 (5.5, 18.0)	13.0 (8.4, 19.3)	.35	13.0 (6.8, 22.9)	13.1 (8.4, 19.9)	.67	13.7 (6.9, 20.6)	12.9 (6.3, 17.7)	.14	14.2 (8.6, 21.1)	12.3 (8.6, 17.3)	.09

^a Limb data are median (minimum, maximum).

^b Indicates a difference between the involved and contralateral limbs of the ACLR group or right and left limbs of the control group at a single time point. Comparisons were made using the exact Wilcoxon signed-rank test.

^c Indicates a difference between the involved limb of the ACLR group at the chosen time and the left leg of the control group. Comparisons were made using the exact Mann-Whitney U test.